WORKED SOLUTION

STEP I 2010, Question 5

Pure • Binomial Coefficient Identities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q5

The key idea

Every one of these sums is the binomial expansion of $(1+x)^n$ in disguise, evaluated at $x = 1$ after we have done something to it. A factor of k in front of $\binom{n}{k}$ is what differentiating x^k produces; a factor $\frac{1}{k+1}$ is what integrating x^k produces. So we differentiate or integrate the identity

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

and then put $x = 1$. For k^2 we need to differentiate twice, with a multiplication by x in between.

Solution

Throughout, n is a positive integer and

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n \qquad (*)$$

holds for every real x. Because both sides are polynomials, we may differentiate or integrate them term by term and the identity survives.

Part (i)

Put $x = 1$ in $(*)$. The left-hand side is 2^n and each x^k becomes 1, so

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n,$$

as required.

Part (ii)

Differentiate $(*)$ with respect to x. The constant term disappears and $\binom{n}{k}x^k$ becomes $k\binom{n}{k}x^{k-1}$:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1}. \qquad (**)$$

Putting $x = 1$ gives

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n\,2^{n-1},$$

as required.

Part (iii)

This time we want $\binom{n}{k}$ divided by $k+1$, which is what we get from $\int_0^1 x^k\,\mathrm{d}x = \frac{1}{k+1}$. Integrating $(*)$ from 0 to 1, the left-hand side is

$$\int_0^1 (1+x)^n\,\mathrm{d}x = \left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \frac{2^{n+1}-1}{n+1},$$

and the right-hand side is

$$\sum_{k=0}^{n}\binom{n}{k}\int_0^1 x^k\,\mathrm{d}x = \sum_{k=0}^{n}\frac{1}{k+1}\binom{n}{k}.$$

Hence

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \cdots + \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}\left(2^{n+1}-1\right),$$

as required. Note that the lower limit matters: integrating from 0 is what produces the -1.

Part (iv)

Ask yourself If we simply differentiate $(**)$ again, we get coefficients $k(k-1)$, not k^2. How can we get k^2 instead?

Before differentiating a second time, multiply $(**)$ by x. That restores the power x^k alongside $k\binom{n}{k}$, so the next derivative brings down another factor of exactly k:

$$nx(1+x)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}x^k.$$

Differentiating, using the product rule on the left,

$$n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2} = \sum_{k=1}^{n} k^2\binom{n}{k}x^{k-1}.$$

(When $n = 1$ the second term on the left has the factor $n - 1 = 0$, so it is simply absent; the identity still holds.) Now put $x = 1$:

$$\begin{aligned}\binom{n}{1} + 2^2\binom{n}{2} + 3^2\binom{n}{3} + \cdots + n^2\binom{n}{n} &= n\,2^{n-1} + n(n-1)2^{n-2} \\ &= n\,2^{n-2}(2 + (n-1)) \\ &= n(n+1)2^{n-2},\end{aligned}$$

as required.

An alternative is to write $k^2 = k(k-1) + k$. Differentiating $(*)$ twice and putting $x = 1$ gives $\sum k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$, and adding the result of part (ii) gives the same answer.

Checks

Take $n = 3$, where the coefficients are $1, 3, 3, 1$.

- (i) $1 + 3 + 3 + 1 = 8 = 2^3$.
- (ii) $3 + 6 + 3 = 12 = 3 \cdot 2^2$.
- (iii) $1 + \frac{3}{2} + 1 + \frac{1}{4} = \frac{15}{4} = \frac{1}{4}(2^4 - 1)$.
- (iv) $3 + 12 + 9 = 24 = 3 \cdot 4 \cdot 2$.

We also checked all four identities exactly, with fractions, for every n from 1 to 40.

Common slips

- In (iii), forgetting the contribution from the lower limit and getting $\frac{2^{n+1}}{n+1}$.
- In (iv), differentiating $(**)$ directly, which gives $\sum k(k-1)\binom{n}{k}$ rather than $\sum k^2\binom{n}{k}$.

Beyond STEP

These identities have probabilistic meanings. If X counts the heads in n tosses of a fair coin, then $\mathrm{P}(X = k) = \binom{n}{k}/2^n$, so part (ii) says $\mathrm{E}(X) = \frac{n}{2}$ and part (iv) says $\mathrm{E}(X^2) = \frac{n(n+1)}{4}$, giving $\mathrm{Var}(X) = \frac{n}{4}$. Differentiating a generating function to extract moments is a standard first-year technique. Grimmett and Welsh, *Probability: An Introduction*, develops it properly.

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